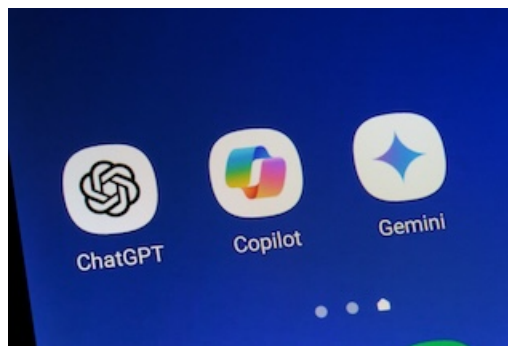


PROJECT: UNDERSTANDING SEARCH INTEREST WITH GOOGLE TRENDS

You are a data analyst working for a tech company closely monitoring the AI tools market. You want to understand the evolving popularity of three major AI tools, ChatGPT, Gemini, and Microsoft Copilot, and identify which tool is gaining the most traction and how they compare over time.



You'll work with real-world time series data that captures the global interest in these tools over several weeks. Your goal is to analyze this data, uncover patterns, and provide actionable insights to help your company make informed decisions. This might include determining where to focus marketing efforts, predicting future trends, or identifying potential areas for improvement.

Are you ready to help your company stay ahead of the curve in the competitive AI tools market? Let's get started!

The Data

The Google Trends data is available as a CSV file `ai_tools_comparison.csv`. The data contains the number of worldwide searches for chatGpt, Gemini, and Microsoft Copilot over the past 12 months as of September 2024.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load the data
trends = pd.read_csv('ai_tools_comparison.csv')

# Inspect the data
trends.head()
```

index	...	↑↓	week	...	↑↓	chatgpt	...	↑↓	gemini	...	↑↓	microsoft_copilot	...
		0	2023-08-27					56			3		
		1	2023-09-03					56			3		
		2	2023-09-10					63			3		
		3	2023-09-17					64			3		
		4	2023-09-24					66			3		

Rows: 5

[↗ Expand](#)

```
# Start coding here
# Use as many cells as you need
```

```
import pandas as pd
import numpy as np

trends = pd.read_csv('ai_tools_comparison.csv')
trends['week'] = pd.to_datetime(trends['week'])

# Calculate slope (linear regression coefficient) for each tool
tools = ['chatgpt', 'gemini', 'microsoft_copilot']
slopes = {}
x = np.arange(len(trends))

for tool in tools:
    slope = np.polyfit(x, trends[tool], 1)[0]
    slopes[tool] = slope

most_consistent_tool = max(slopes, key=slopes.get)
print(f"Slopes: {slopes}")
print(f"most_consistent_tool = '{most_consistent_tool}'")

Slopes: {'chatgpt': 0.3339813226605672, 'gemini': 0.2854964741757194, 'microsoft_copilot': 0.06971602820659421}
most_consistent_tool = 'chatgpt'
```

```
import matplotlib.pyplot as plt
import matplotlib.dates as mdates

fig, ax = plt.subplots(figsize=(12, 5))

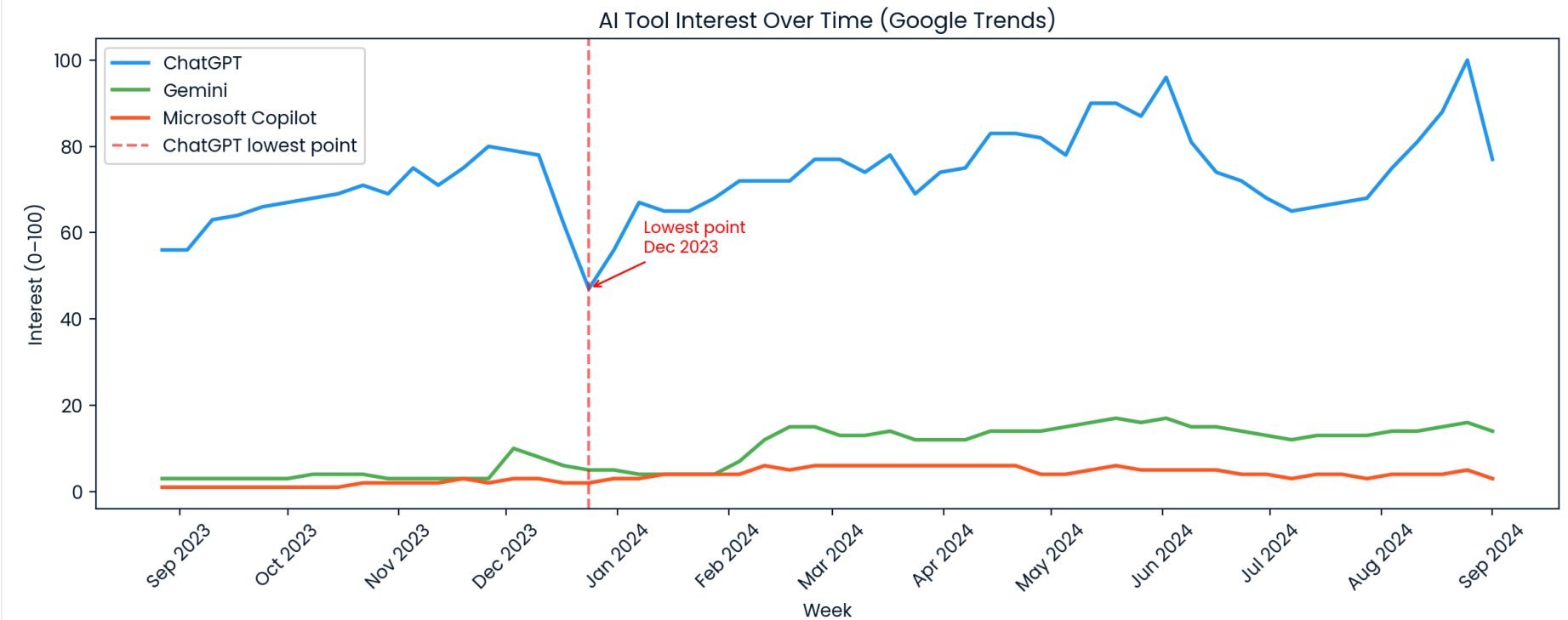
ax.plot(trends['week'], trends['chatgpt'], label='ChatGPT', color='#2196F3', linewidth=2)
ax.plot(trends['week'], trends['gemini'], label='Gemini', color='#4CAF50', linewidth=2)
ax.plot(trends['week'], trends['microsoft_copilot'], label='Microsoft Copilot', color='#FF5722', linewidth=2)

# Find the week where ChatGPT interest was at its LOWEST value
dip_idx = trends['chatgpt'].idxmin()
dip_date = trends.loc[dip_idx, 'week']

ax.axvline(x=dip_date, color='red', linestyle='--', alpha=0.6, label=f'ChatGPT lowest point')
ax.annotate(f"Lowest point\n{dip_date.strftime('%b %Y')}",
            xy=(dip_date, trends.loc[dip_idx, 'chatgpt']),
            xytext=(30, 20), textcoords='offset points',
            arrowprops=dict(arrowstyle='->', color='red'),
            color='red', fontsize=9)

ax.xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))
ax.xaxis.set_major_locator(mdates.MonthLocator(interval=1))
plt.xticks(rotation=45)
ax.set_title('AI Tool Interest Over Time (Google Trends)')
ax.set_ylabel('Interest (0-100)')
ax.set_xlabel('Week')
ax.legend()
plt.tight_layout()
plt.savefig('ai_tools_interest.png', dpi=150)
plt.show()

# Save as the month/year of ChatGPT's lowest interest value
gpt_dip = dip_date.strftime('%B %Y')
print(f"gpt_dip = '{gpt_dip}'")
```



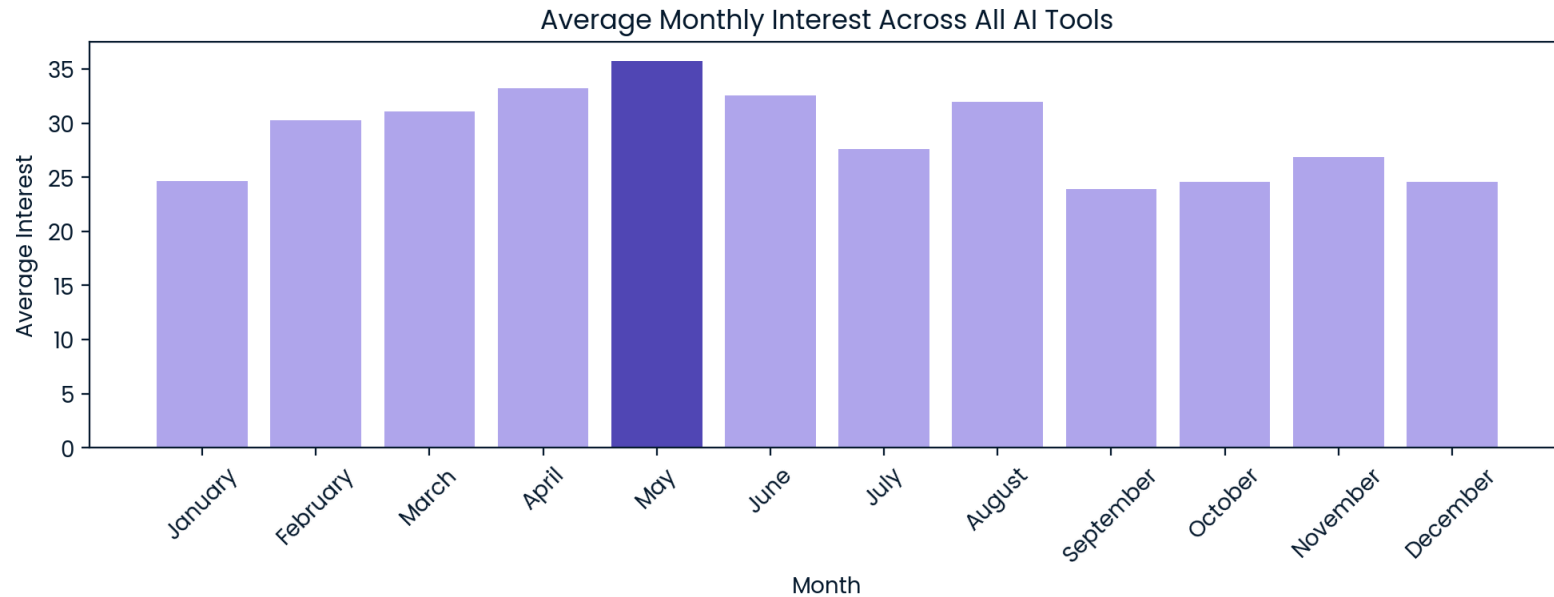
```
gpt_dip = 'December 2023'
```

```
# Extract month name and compute average interest across all tools
trends['month'] = trends['week'].dt.strftime('%B')
trends['month_num'] = trends['week'].dt.month
trends['avg_interest'] = trends[['chatgpt', 'gemini', 'microsoft_copilot']].mean(axis=1)

monthly_avg = (
    trends.groupby(['month_num', 'month'])['avg_interest']
    .mean()
    .reset_index()
    .sort_values('month_num')
)

fig, ax = plt.subplots(figsize=(10, 4))
colors = ['#534AB7' if m == monthly_avg.loc[monthly_avg['avg_interest'].idxmax(), 'month']
          else '#AFA9EC' for m in monthly_avg['month']]
ax.bar(monthly_avg['month'], monthly_avg['avg_interest'], color=colors)
ax.set_title('Average Monthly Interest Across All AI Tools')
ax.set_ylabel('Average Interest')
ax.set_xlabel('Month')
plt.xticks(rotation=45)
plt.tight_layout()
plt.savefig('seasonality.png', dpi=150)
plt.show()

best_month = monthly_avg.loc[monthly_avg['avg_interest'].idxmax(), 'month']
print(f"best_month = '{best_month}'")
```



```
best_month = 'May'
```

```
print(f"most_consistent_tool = '{most_consistent_tool}'")  
print(f"gpt_dip = '{gpt_dip}'")  
print(f"best_month = '{best_month}'")
```

```
most_consistent_tool = 'chatgpt'  
gpt_dip = 'December 2023'  
best_month = 'May'
```